AULA 6

Nome

Curso FICHA PRÁTICA Nº 3

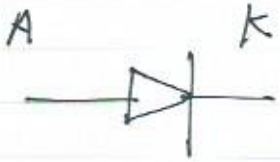
Prova Escrita de:

N.º Aluno

Ano Letivo / Data da Avaliação

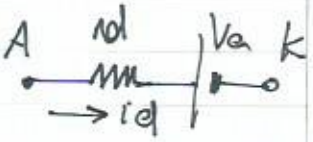
N.º Folhas (1º)

Época:



$\Rightarrow$

- Díodo conduz $V_{AK} > V_\alpha$



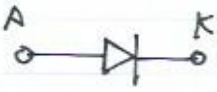
- Díodo n/ conduz $V_{AK} < V_\alpha$

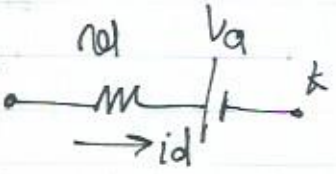


$i_d = 0$

ANÁLISE CIRCUITO COM UM DÍODO.

(1.) ANALISAR ESTADO DE CONDUÇÃO

a) SUBSTITUIR 

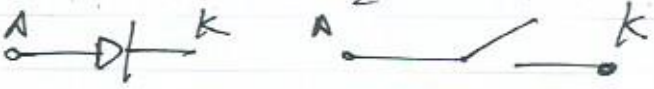
$\Rightarrow$ 

b) Impor sentido que a corrente $a \rightarrow k$

c) Impor condição $i_d > 0 \Rightarrow$ Identificar qual valor V_{in} diodo conduz

d) OBTER RELAÇÃO $V_{out} = f(V_{in})$ $\rightarrow$ ESTADO DE CONDUÇÃO.

(2.) ANALISAR ESTADO DE N/ CONDUÇÃO.

a) SUBSTITUIR 

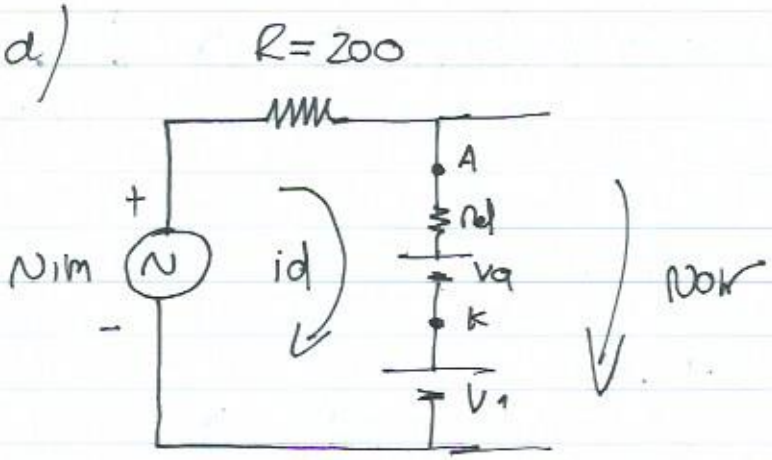
b) Condição deste estado = INVERSA DO ESTADO CONDUÇÃO.

c) CALCULAR RELAÇÃO $V_{out} = f(V_{in})$

(3-) OBTER FUNÇÃO TRANSFERÊNCIA ANALÍTICA

Juntar numa só EQUAÇÃO AMBOS OS ESTADOS (ANÁLISE)

1b) a)



$$-V_{im} + R \times id + Rd \times id + Va + V_1 = 0$$

$$id = \frac{V_{im} - Va - V_1}{R + Rd} > 0 \Rightarrow V_{im} > Va + V_1$$

$$V_{im} > 0.7 + 1 = 1.7$$

Díodo conduz quando $V_{im} > 1.7V$

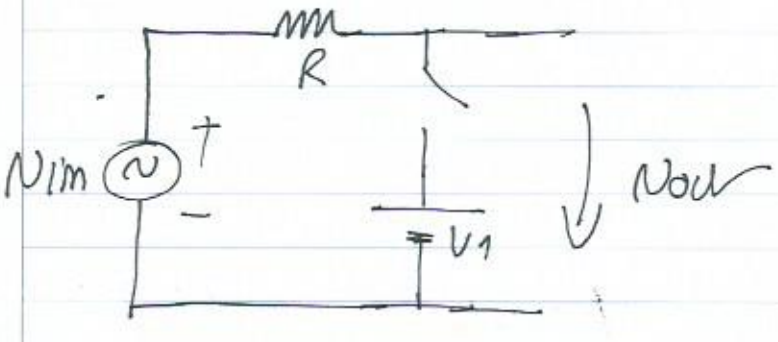
b)

ESTADO DE CONDUÇÃO ($V_{im} \geq 1,7$)

$$V_{OUT} = Rd \times id + Va + V_1 = Rd \times \left(\frac{V_{im} - Va - V_1}{R + Rd} \right) + Va + V_1$$

$$V_{OUT} = \frac{100}{200 + 100} \times (V_{im} - 1.7) + 1.7 \quad , V_{im} > 1.7$$

ESTADO DO N/ conexão:



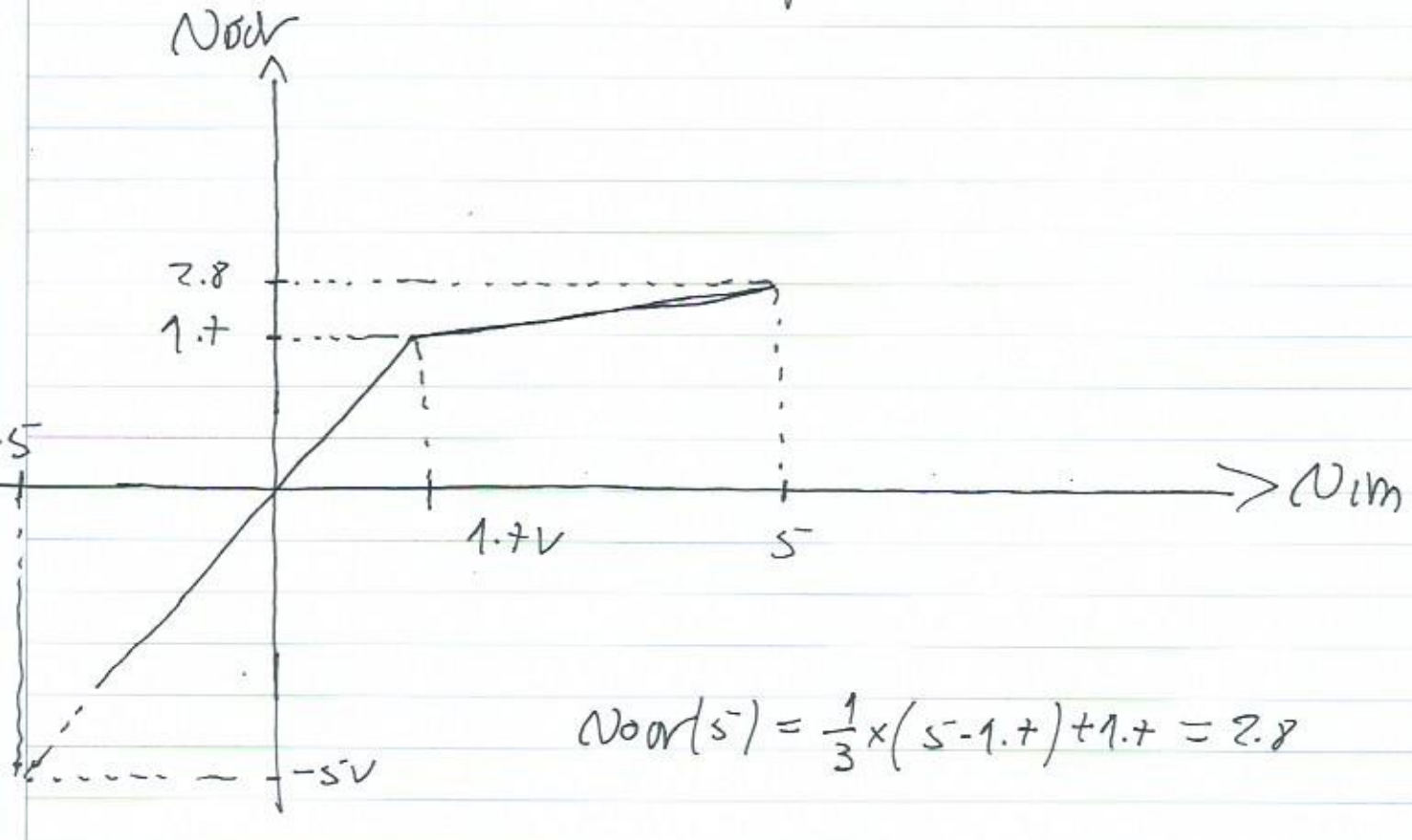
$V_{out} - V_{im} + V_d = 0$

$i_d = 0$

$\Rightarrow \quad V_{out} = V_{im}$

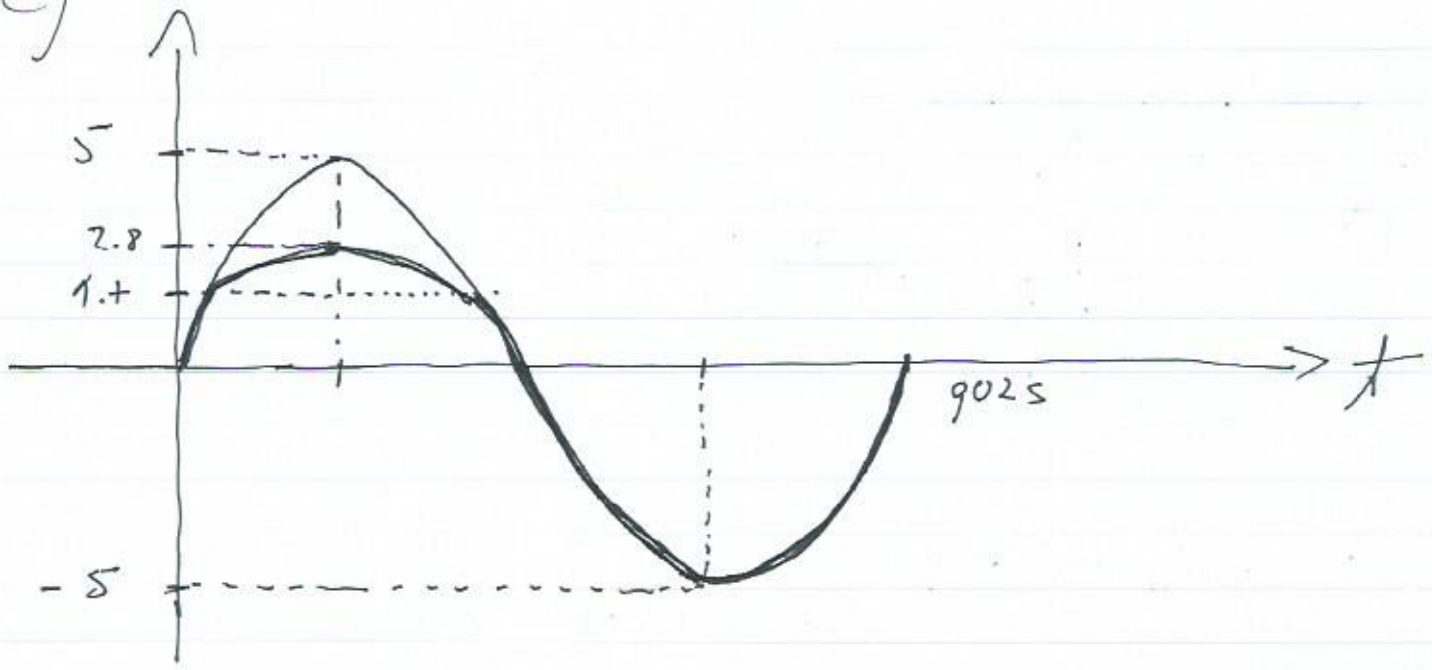
FUNÇÃO TRANSFORÊNCIA ANALÍTICA

$$V_{out}(V_{im}) = \begin{cases} \frac{1}{3} \times (V_{im} - 1.7) + 1.7, & V_{im} \geq 1.7V \\ V_{im}, & V_{im} \leq 1.7V \end{cases}$$



$$V_{out}(5^-) = \frac{1}{3} \times (5^- - 1.7) + 1.7 = 2.8$$

c)



d) SIMULAÇÃO PSPICE

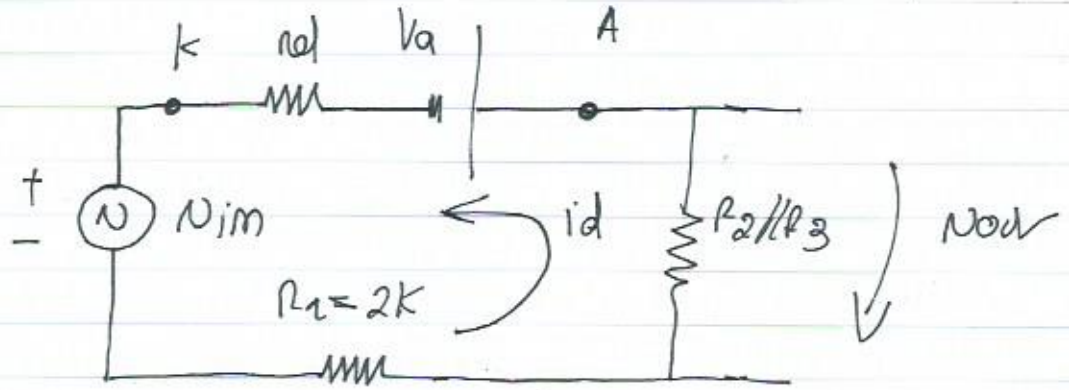
e) SIMULAÇÃO PYTHON

17/

$V_d = 0.7V$

$R_d = 100\Omega$

a) $v_{im} = ?$ $v_{out} = ?$ $i_d = 1mA \Rightarrow$ díodo conduz.



$$v_{im} + R_1 \times id + R_2 // R_3 \times id + V_a + r_d \times id = 0$$

$$id = \frac{-v_{im} - V_a}{R_1 + R_2 // R_3 + r_d} > 0 \Rightarrow -v_{im} - V_a > 0 \Rightarrow v_{im} < -V_a$$

Díodo conduz quando $v_{im} < -0,7V$

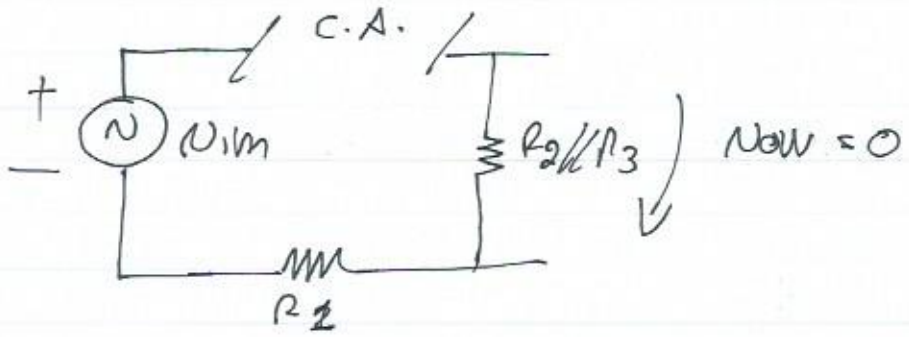
b) ESTADO DO CONDUTOR:

$$v_{out} = R_2 // R_3 \times (-id) = R_2 // R_3 \times \left(\frac{v_{im} + V_a}{R_1 + R_2 // R_3 + r_d} \right)$$

$$= \frac{500 \times (v_{im} + 0.7)}{2K + 500 + 100} = \frac{500}{2600} \times (v_{im} + 0.7)$$

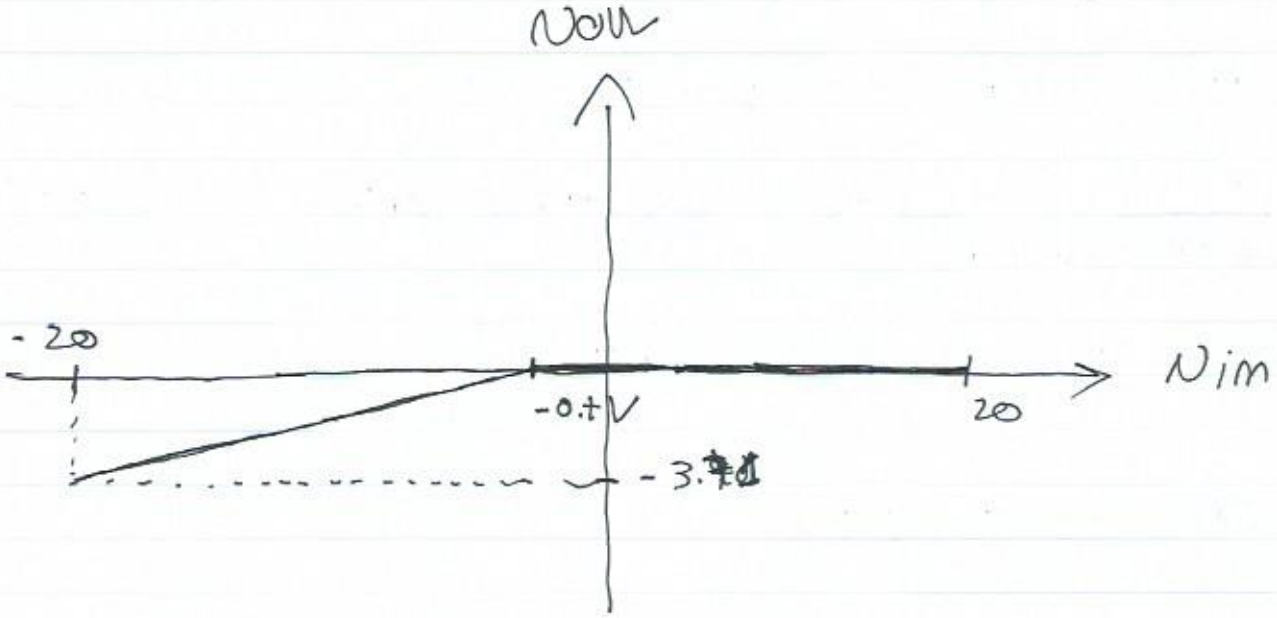
$$= \frac{5}{26} \times (v_{im} + 0.7) \quad , v_{im} \leq -0.7V$$

ESTADO DO 1º CONDUÇÃO.



FUNÇÃO TRANSFERÊNCIA ANALÍTICA

$$V_{out} = \begin{cases} 0 & , V_{in} \geq -0.7\,V \\ (V_{in} + 0.7) \times \frac{5}{26} & , V_{in} \leq -0.7\,V \end{cases}$$



$$V_{out}(-20V) = \frac{(-20 + 0.7) \times 5}{26} \cong -3.9\,V$$

Nome

N.º Aluno

Curso

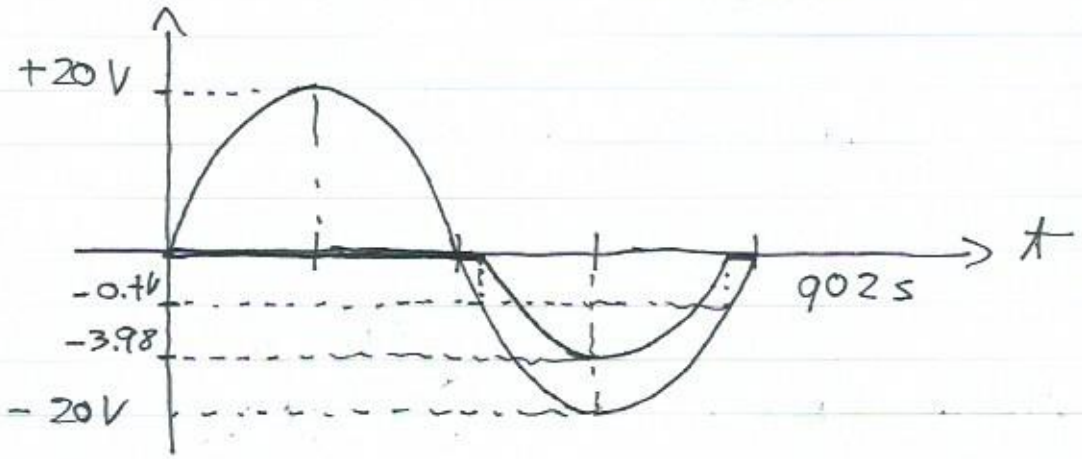
Ano Letivo / Data da Avaliação /

Prova Escrita de:

N.º Folhas (7)

Época:

c)

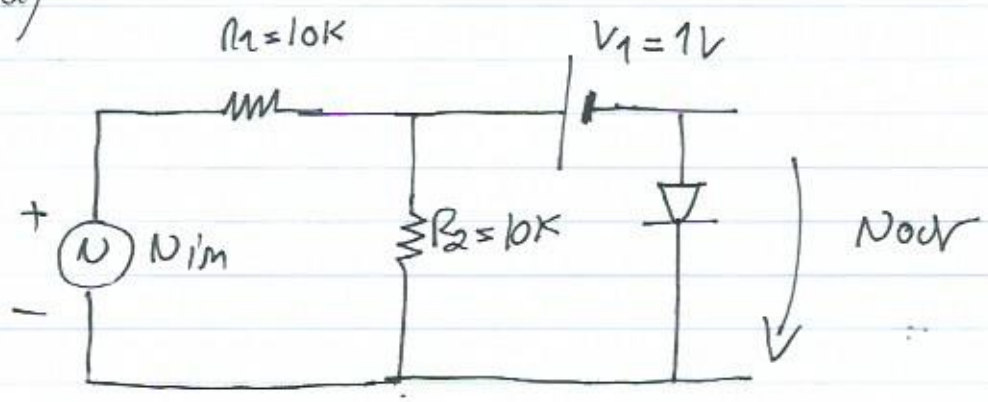


d) Simulação PSPICE.

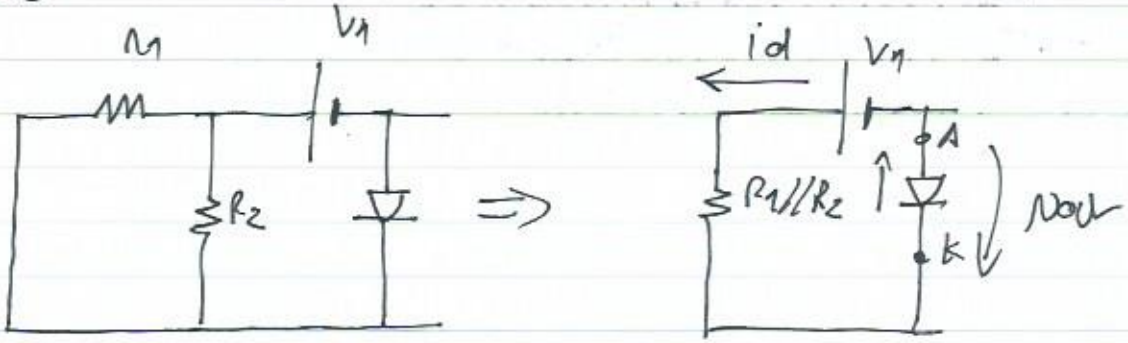
e) Simulação Python.

1/ a)

$V_d = 0.7V$
$r_d = 100\Omega$

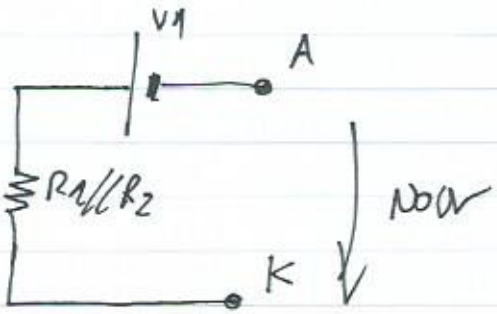


$V_{in} = 0$



Diodo in / conduce

Corrent K → A (INVERSAMENTE POLARIZZATO)



$V_{out} + R_1 // R_2 \cdot i_d + V_1 \leq 0$

$i_d = 0$

$\Rightarrow \quad V_{out} = -V_1 = -1V$

Nome

N.º Aluno

Curso

Ano Letivo / Data da Avaliação / /

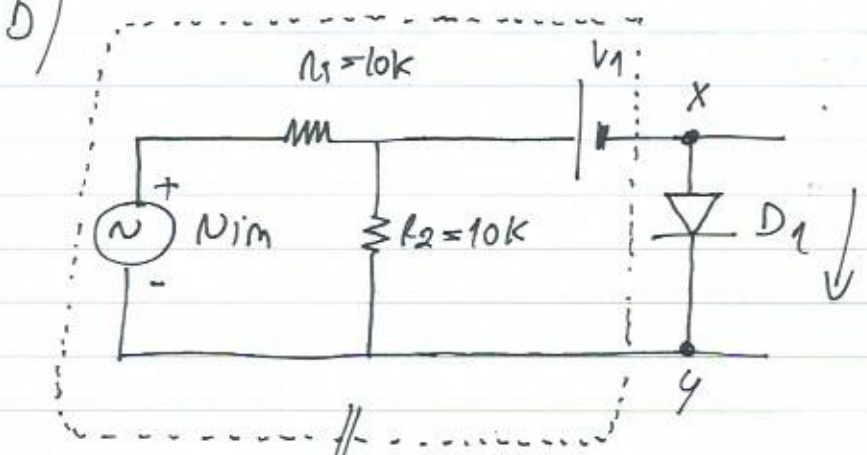
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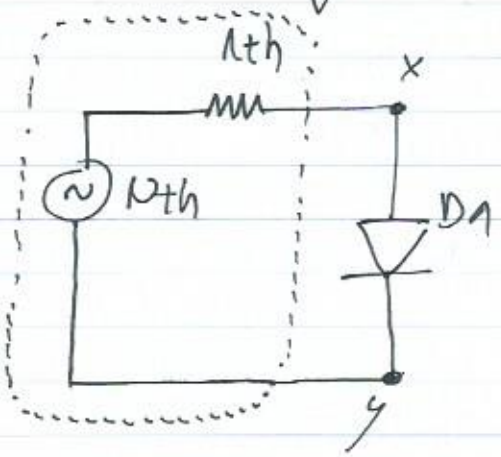
N.º Folhas

Época:

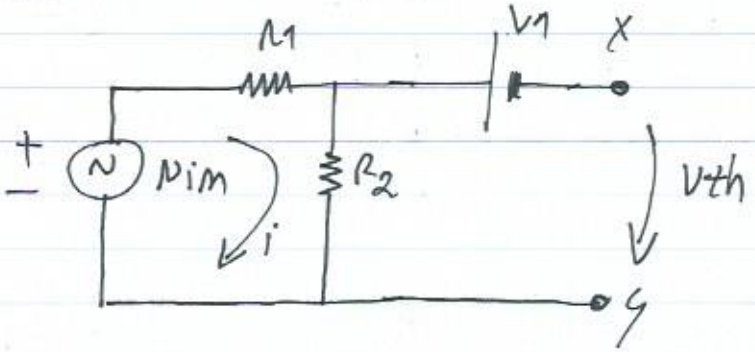
9

b)





1º) Calculo Vth

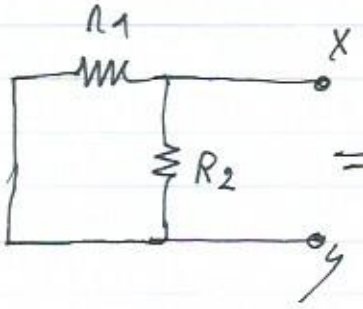


$$-V_{im} + (R_1 + R_2)\,i = 0 \Rightarrow i = \frac{V_{im}}{R_1 + R_2}$$

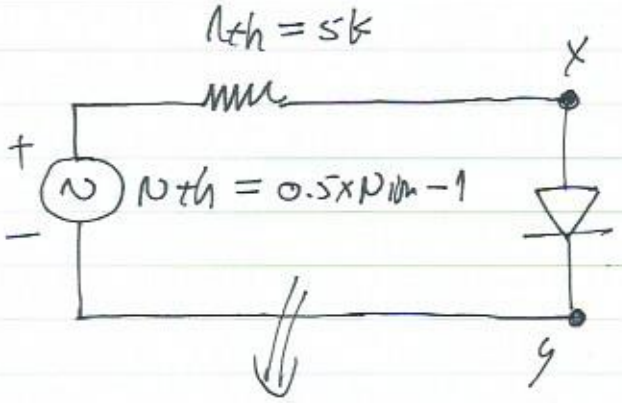
$V_{th} = ?$ $\quad V_{th} + R_2 \times (-i) + V_1 = 0$

$$V_{th} = R_2 \times i - V_1$$

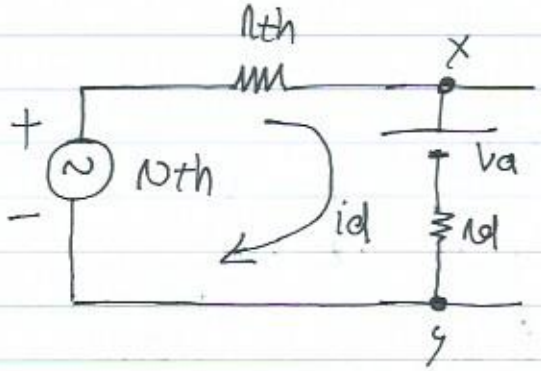
$$V_{th} = \frac{R_2 \times V_{im}}{R_1 + R_2} - V_1 = \frac{10k \times V_{im}}{10k + 10k} - 1 = +0.5 \times V_{im} - 1$$



$\Rightarrow$ $R_{th} = R_1 // R_2 = 10k // 10k = \underline{\underline{5k}}$



ESTADO I – CONDUZ



$$-V_{th} + R_{th} \times i_d + V_a + R_d \times i_d = 0$$

$$i_d = \frac{V_{th} - V_a}{R_{th} + R_d} > 0$$

$$V_{th} > V_a$$

Díodo conduz quando $V_{th} > V_a$

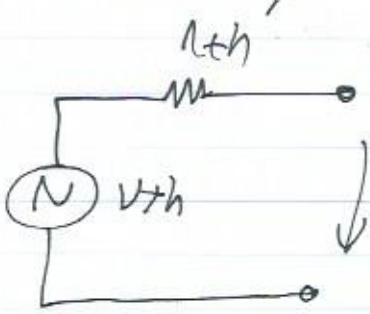
$0.5 \times V_{im} - 1 > V_a \Rightarrow 0.5 \times V_{im} - 1 > 0.7 \Rightarrow 0.5 \times V_{im} > 1.7$

$V_{im} > 3.4\,V \Rightarrow$ Díodo conduz

$$V_{out} = V_a + R_d \times i_d = 0.7 + 100 \times \left(\frac{V_{th} - 0.7}{R_{th} + R_d} \right)$$

$$= 0.7 + \frac{100}{5100} \times (V_{th} - 0.7) \quad , \; \underline{\underline{V_{im} \geq 3.4\,V}}$$

Diodo n/ corrente



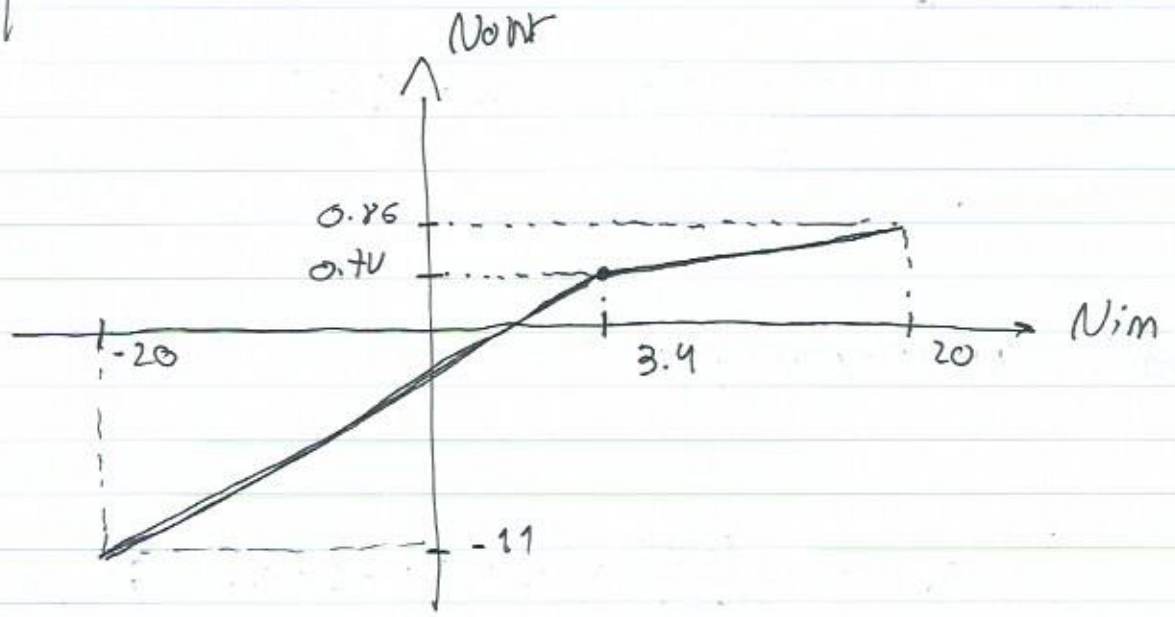
$V_{OUT} = V_{th} = 0.5 \times V_{in-1}$

FUNÇÃO TRANSFERÊNCIA AMPLIFICA

$$V_{OUT} = \begin{cases} 0.7 + \frac{100}{5100} \times (0.5 \times V_{in-1} - 0.7) & ; V_{in} \geq 3.4 \\ 0.5 \times V_{in-1} & ; V_{in} \leq 3.4 \end{cases}$$

$$V_{OUT} = \begin{cases} 0.7 + \frac{100}{5100} \times (0.5 \times V_{in-1} - 1.7) & ; V_{in} \geq 3.4 \\ 0,5\, V_{in-1} & ; V_{in} \leq 3.4 \end{cases}$$

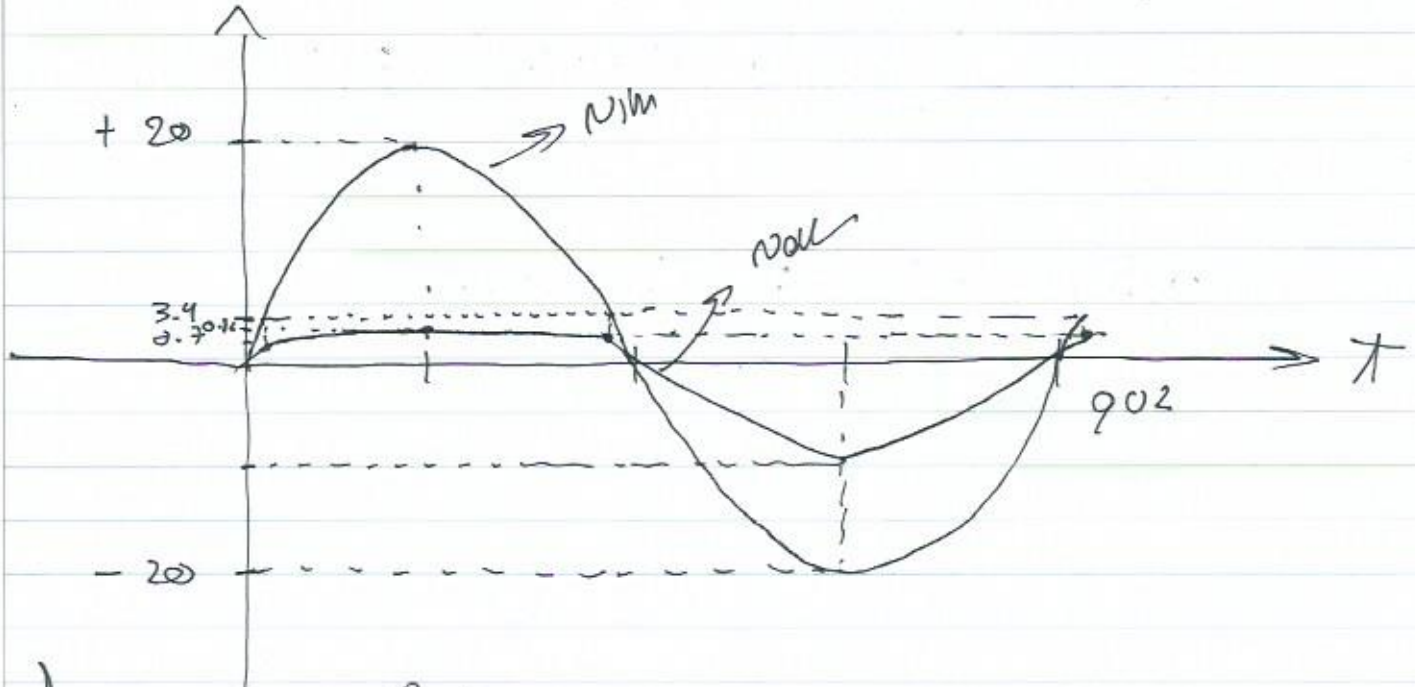
$$V_{out} = \begin{cases} 0.7 + \frac{1}{5\Omega}\times(0.5\,V_{im} - 1.7) & ; V_{im} \geq 3.4V \\ 9.5 \times V_{im} - 1 & ; V_{im} \leq 3.4 \end{cases}$$



$V_{out}(+20) = 0.7 + \frac{1}{5\Omega}\times(0.5\times 20 - 1.7) = 9.1627 + 0.7 = 0.86V$

$V_{out}(-20) = 9.5\times(-20) - 1 = -11$

$V_{out}(+3.4) = 0.7V$



d) SIMULAÇÃO PSPICE

e) " PYTHON